Trainings

General Information

The oil filter head contains two spring loaded plungers. One plunger controls the oil pressure for the piston cooling nozzles. The second spring plunger will bypass oil if a filter element becomes plugged or clogged.

When installing a new filter element, always check to be sure there is no interference between the filter head adapter and the element.

The oil filter remote option has a transfer connection and a remote filter head. The transfer connection is attached to the block in the same manner as a standard filter head. The transfer connection also houses the pressure regulator plunger for the piston cooling nozzles. The remote filter head houses the pressure regulator plunger for the filter bypass.

Note : The transfer connection gasket is the same as the filter head gasket, but the tab on the end of the gasket must be removed.

Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

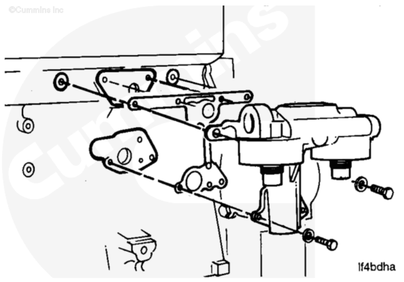
- Drain the engine oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/20/20-007-

037.html)
- Remove the oil filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html)

Remove

Remove the six capscrews and the filter head assembly.

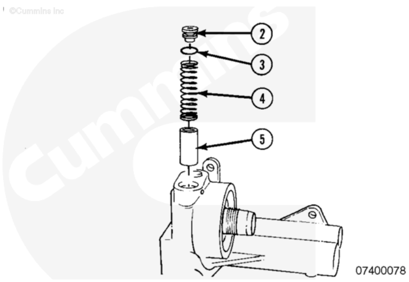
Remove and discard the gasket.



Disassemble

⚠ WARNING ⚠

The threaded plug is under pressure, wear goggles to reduce the possibility of personal injury.



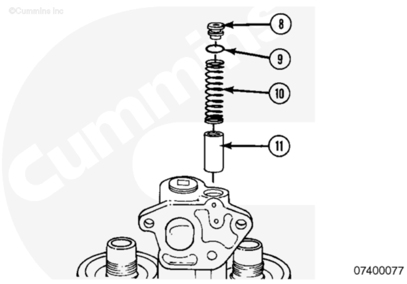
Remove the bypass regulator plunger and related parts.

1. Regulator plug
2. O-ring
3. Bypass valve spring
4. Bypass valve regulator spring

Discard the o-ring.

⚠ WARNING ⚠

The threaded plug is under pressure, wear goggles to reduce the possibility of personal injury.



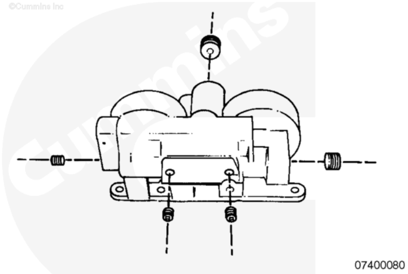
Remove the piston cooling nozzle plunger and related parts.

1. Pressure regulator plug
2. O-ring

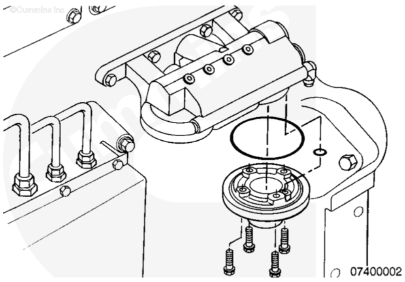
3. Piston cooling valve spring
4. Piston cooling valve plunger

Discard the o-ring.

Remove the remaining straight threaded o-ring plugs.



Remove the filter head adapter. Refer to Procedure 007-018 in Section 7. (/qs3/pubsys2/xml/en/procedures/20/20-007-018.html)



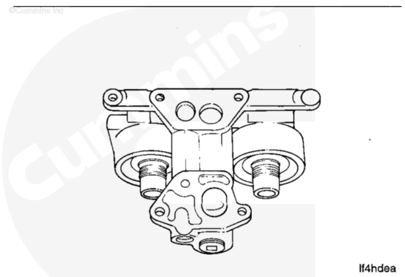
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use a gasket scraper to remove the two mounting gaskets.

Use solvent to clean the filter head.

Dry with compressed air.

Inspect the part for cracks or other damage.

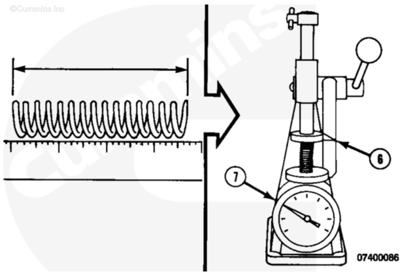
Check the visible portion of the valve plungers for signs of scoring or other damage.

If any of the plungers are damaged, the filter head **must** be rebuilt.

Refer to Procedure 007-015 in Section 7. (/qs3/pubsys2/xml/en/procedures/20/20-007-015-tr.html)

Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-046-tr.html)

Check the bypass plunger spring.

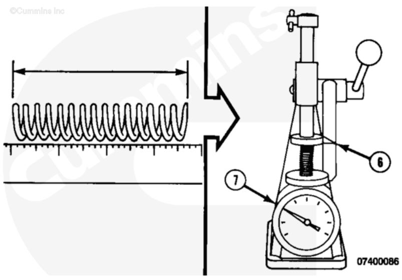


Bypass Valve and Piston Cooling Regulation Spring				
	mm		in	
Free Length	88.98	MIN	3.500	
Working Height (6)		50.80	MAX	2.000

Use a valve spring tester, Part Number 3375182, or equivalent to measure the spring force (7) and the working height (6).

Spring Force (7)		
n		lbf
86	MIN	19.3
95	MAX	21.3

Check the piston cooling nozzle spring.



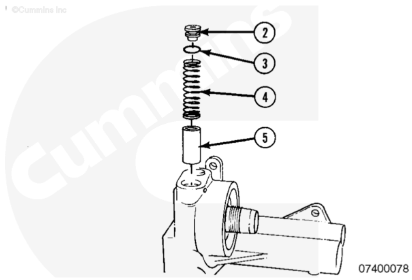
Piston Cooling Nozzle Spring				
	mm		in	
Free Length	88.98	MIN	3.500	
Working Height (6)		50.80	MAX	2.000

Use a valve spring tester, Part Number 3375182, or equivalent to measure the spring force (7) and the working height (6).

Spring Force (7)		
n.m		ft-lb
26	MIN	19.3
29	MAX	21.3

Assemble

Use clean engine oil to lubricate the bypass regulator plunger and related parts.



Install the new o-ring (3) on the regulator plug (2).

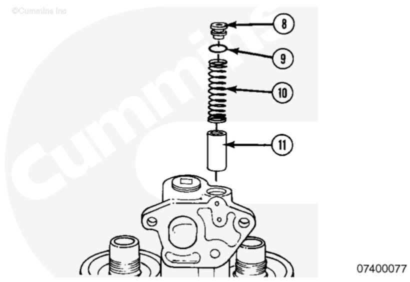
Install the parts.

1. Bypass valve regulator plunger
1. Bypass valve spring
1. Regulator plug with o-ring (3)

Tighten the regulator plug.

Torque Value: 54 n•m [40 ft-lb]

Use clean engine oil to lubricate the piston cooling nozzle plunger and related parts.



Install the new o-ring (9) on the piston cooling nozzle plunger plug (8).

Install the parts.

1. Piston cooling valve plunger
1. Piston cooling valve spring
1. Pressure regulator plug with o-ring (9)

Tighten the piston cooling nozzle plunger plug.

Torque Value: 54 n•m [40 ft-lb]

Tighten the straight threaded o-ringed plugs.

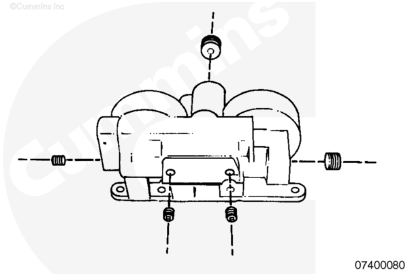
Torque Value:
1-5/16

1. 61 n•m [45 ft-lb]

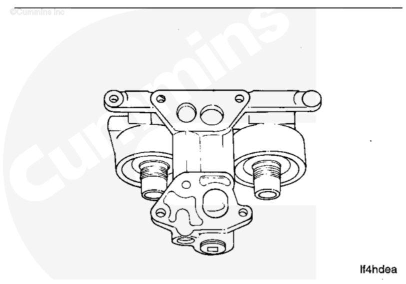
Torque Value:

9/16

1. 14 n•m [124 in-lb]



Install the filter head adapter. Refer to Procedure 007-018 in Section (/qs3/pubsys2/xml/en/procedures/20/20-007-018.html)

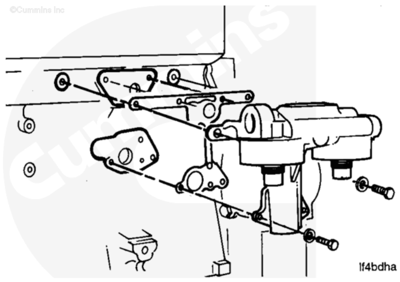


Install

Install the gasket, filter head and six capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

⚠ CAUTION ⚠

Use caution when filling the cooling system so that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations

- Install new combination oil filters. Refer to Procedure 007-013 in Section 7.

(/qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html)

- Fill the engine with clean, 15W-40 oil. Refer to Procedure 007-037 in Section 7 (/qs3/pubsys2/xml/en/procedures/20/20-007-037.html) for engine oil capacity.
- Operate the engine to normal operating temperature and check for leaks.

Last Modified: 20-Aug-2015